

STUDENT NUMBER: _____

STUDENT NAME: _____



THE HILLS GRAMMAR SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

2008

MATHEMATICS

EXTENSION 1

Teacher Responsible: Mrs P Singh

General Instructions:

- Reading time - 5 minutes
- Working time - 2 hours
- This paper contains 7 questions.
- ALL questions to be attempted.
- ALL questions are of equal value.
- ALL necessary working should be shown in every question in the booklets provided.
- **Start each question in a new booklet.**
- A table of standard integrals is supplied at the back of this paper.
- Marks may be deducted for careless, untidy or badly arranged work.
- Board approved calculators may be used.
- Hand up your paper in ONE bundle, together with this question paper.

Students are advised that this is a Trial Examination only and cannot in any way guarantee the content or the format of the Higher School Certificate Examination.

Trial Extension 1 – 2008**Question 1 (12 marks)****Marks**

- a) The polynomial $P(x) = x^3 + ax^2 + 2x - 4$ has a remainder of -7 when divided by $x + 2$. Find the value of a . **1**
- b) Differentiate $e^{2x} \sin x$ **2**
- c) Find the acute angle, to the nearest degree, between the lines
 $2x + y = 4$ and $x - y = 2$ **2**
- d) Evaluate $\int_0^2 \frac{dx}{4+x^2}$ **2**
- e) Using the substitution $u = 2x + 1$ or otherwise, find $\int_0^1 \frac{4x}{2x+1} dx$. **3**
- f) Find the co-ordinates of the point P which divides the line joining $A(-3, 4)$ and $B(2, -8)$ externally in the ratio $2:5$. **2**

Question 2 (12 marks)

- a) Prove the identity: $\frac{\cos x - \cos 2x}{\sin 2x + \sin x} = \operatorname{cosec} x - \cot x$ **3**
- b) Find all angles for θ , where $0 \leq \theta \leq 2\pi$ for which $\sqrt{3} \cos \theta - \sin \theta = 1$. **4**
- c) The function $h(x)$ is given by $h(x) = \sin^{-1} x + \cos^{-1}(x)$ for $-1 \leq x \leq 1$.
- i) show that $h'(x) = 0$ **1**
- ii) sketch the graph of $y = h(x)$ **2**
- d) Find $\frac{dy}{dx}$ if $y = \tan^{-1}(\sin x)$ **2**

Question 3 (12 marks)**Marks**

- a) A cup of hot cappuccino at temperature T^0 Celsius loses heat when placed in a cooler environment. It cools according to the law

$$\frac{dT}{dt} = -k(T - T_0) \text{ where time, } t \text{ is the time elapsed in minutes and}$$

T_0 is the temperature of the environment in degrees Celsius.

- i) Show that $T = T_0 + Ce^{-kt}$ 1

- ii) A cup of cappuccino at 100^0C is placed in an environment at -20^0C for 3 minutes and then cools to 70^0C . Find k , in exact form. 2

- iii) The same cup of cappuccino at 70^0C is then placed in an environment at 20^0C , assuming k stays the same, find the temperature, to the nearest degree, of the cappuccino after a further 15 minutes. 4

- b) i) Show that $x = 2$ is a zero of $x^3 - 4x^2 + 8$ 1

- ii) Hence find all the real zeros of $x^3 - 4x^2 + 8$, leaving your answers in exact form. 2

- iii) Hence solve the inequality: $\frac{4}{x-2} \leq x$ 2

Question 4 (12 marks)**Marks**

- a) Solve $2^{2x+1} - 5(2^x) + 2 = 0$ **3**
- b) Find the coefficient of x^3 in $\left(3x^2 + \frac{1}{x}\right)^9$ **3**
- c) A function is defined as $f(x) = 1 + e^{2x}$.
- i) Write down the range of $f(x)$. **1**
- ii) Given $f^{-1}(x)$ is the inverse function for $f(x)$, show that
- $$f^{-1}(x) = \frac{1}{2} \ln(x-1)$$
- 2**
- iii) On the same set of axes, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$, showing all key features. **3**

Question 5 (12 marks)**Marks**

- a) A particle moves in a straight line with Simple Harmonic Motion. At time t seconds, its displacement x metres from a fixed point O , is given by:

$$x = 5 \sin \frac{\pi}{2} \left(t + \frac{1}{3} \right)$$

- i) Show that $\ddot{x} = -\frac{\pi^2}{4} x$ 2
- ii) State the period and the amplitude of the motion. 2

- b) The acceleration of a particle moving in a straight line is given by:

$$\frac{d^2x}{dt^2} = \frac{-72}{x^2},$$

where x metres is the displacement from the origin after t seconds. Initially the particle is 9 metres to the right of the origin with a velocity of 4m per second.

- i) Show that the velocity v of the particle in terms of x is $v = \frac{12}{\sqrt{x}}$.
Explain why v is always positive for the given initial conditions. 5
- ii) Find an expression for t in terms of x . 3

Question 6 (12 marks)

Marks

- a) i) Show that the equation of the tangent to the parabola $x^2 = 16y$ at any point $P(8t, 4t^2)$ on it is $y = tx - 4t^2$.

2

- ii) Show that the equation of the line r through the focus S of the parabola which is perpendicular to the focal chord through P is

$$(t^2 - 1)y + 2tx = 4(t^2 - 1)$$

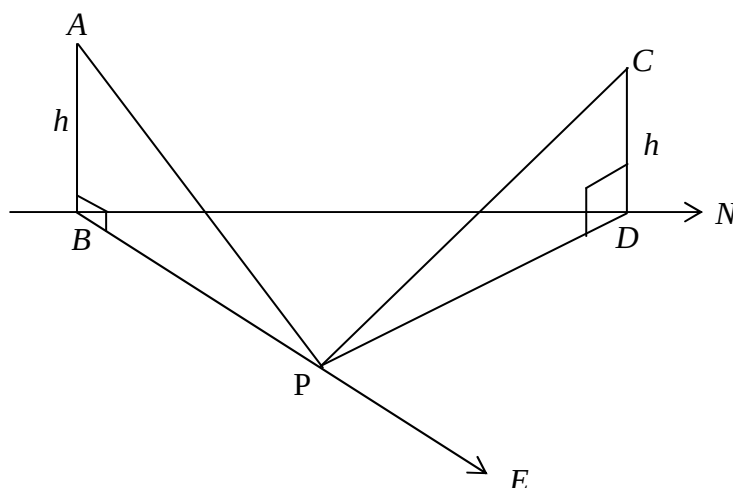
2

- iii) Prove that the locus of the point of intersection of the line r and the tangent at P is a horizontal line.

3

- b) AB and CD are two towers of equal height (h). CD is due north of AB . From a point P on the same horizontal plane as the feet B and D of the towers, and bearing due east of the tower AB , the angles of elevation of A and C , the tops of the towers, are 47° and 31° respectively. If the distance between the towers is 88m, find the height of the towers to the nearest metre.

5



Question 7 (12 marks)**Marks**

- a) Prove by mathematical induction that
 $1.2^0 + 2.2^1 + 3.2^2 + \dots + n.2^{n-1} = 1 + (n-1)2^n$ for $n \geq 1$. **4**
- b) During a soccer tournament, Juan is standing 25m away from the goal line. He kicks a soccer ball off the ground at an angle of 30° to the horizontal with an initial velocity of V m per sec. The ball hits the top bar which is 2.4 m directly above the goal line. Neglecting air resistance and assuming that acceleration due to gravity is 10 m/s^2 , find:
- i) The horizontal and vertical components of displacement of the ball in terms of the initial velocity V . **4**
- ii) The Cartesian equation of the motion for the path of the ball. **1**
- iii) The initial velocity of the ball, correct to 1 decimal place. **3**

END OF EXAMINATION

THCS Solen Trial 2008 Ext 1

Q1 (12 marks)

$$(a) -8 + 4a - 4 - 4 = -7$$

$$4a = 9$$

$$\therefore a = \frac{9}{4}$$

(1)

$$(b) y = e^{2x} \sin x$$

$$y' = e^{2x} \cos x + \sin x \cdot 2e^{2x}$$

$$= e^{2x} (\cos x + 2 \sin x)$$

(2)

$$(c) y = -2x + 4 \quad \therefore m_1 = -2$$

$$y = x - 2 \quad m_2 = 1$$

$$\therefore \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = |3|$$

$$\therefore \theta = 72^\circ \quad (2)$$

$$(d) \int_0^2 \frac{dx}{4+x^2} = \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2$$

$$= \frac{1}{2} \tan^{-1}(1) - \frac{1}{2} \tan^{-1}(0)$$

$$= \frac{1}{2} \times \frac{\pi}{4} - \frac{1}{2} \times 0$$

$$= \frac{\pi}{8}$$

(2)

$$(e) \int_0^1 \frac{4x}{2x+1} dx$$

$$u = 2x + 1 \Rightarrow 2x = u - 1$$

$$du = 2dx$$

$$\text{when } x=1, \quad u=3$$

$$x=0, \quad u=1$$

$$= \int_1^3 \frac{u-1}{u} du$$

$$= \int_1^3 \left(1 - \frac{1}{u} \right) du$$

$$= [u - \log_e u]_1^3$$

$$= (3 - \log_e 3) - (1 - \log_e 1)$$

$$= 2 - \log_e 3 \quad (3)$$

$$(f) P \left(\frac{-2 \times 2 + 5 \times 3}{-2 + 5}, \frac{-2 \times 8 + 5 \times 4}{-2 + 5} \right)$$

$$= \left(-\frac{19}{3}, \frac{36}{3} \right)$$

$$= \left(-6\frac{1}{3}, 12 \right)$$

(2)

Q2. 12 marks

$$(a) \text{ LHS} = \frac{\cos x - \cos 2x}{\sin 2x + \sin x} = \frac{\cos x + 2\cos^2 x + 1}{2\sin x \cos x + \sin x}$$

$$= \frac{(1 + 2\cos x)(1 - \cos x)}{\sin x (2\cos x + 1)}$$

$$= \frac{1 - \cos x}{\sin x}$$

$$= \frac{1}{\sin x} - \frac{\cos x}{\sin x}$$

$$= \csc x - \cot x$$

QED

(3)

$$(b) \sqrt{3} \cos \theta - \sin \theta = r \cos (\theta + \alpha)$$

$$= r \cos \theta \cos \alpha - r \sin \theta \sin \alpha$$

$$r \cos \theta = \sqrt{3}$$

$$r \sin \theta = 1$$

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$r^2 = (\sqrt{3})^2 + 1^2$$

$$r = 2$$

$$\therefore 2 \cos \left(\theta + \frac{\pi}{6} \right) = 1$$

$$\cos \left(\theta + \frac{\pi}{6} \right) = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{3} - \frac{\pi}{6}, \quad \frac{5\pi}{3} - \frac{\pi}{6}$$

$$= \frac{\pi}{6} \text{ or } \frac{3\pi}{2}$$

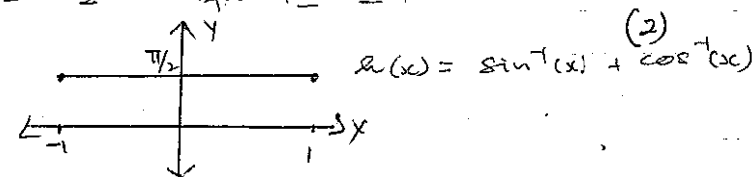
(4)

$$(c) (i) f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0 \quad \text{as reqd.} \quad (1)$$

(ii) $\therefore$ function $f(x)$ has a gradient of zero
 $\therefore$ function is a straight line of the form
 $y = a$ where a is the constant.

$$\text{when } x=0, \quad f(x) = \frac{\pi}{2}$$

$$\therefore f(x) = \frac{\pi}{2} \quad \text{for } -1 \leq x \leq 1$$



$$(d) \text{ Let } u = \sin x$$

$$\frac{du}{dx} = \cos x$$

$$\therefore y = \tan^{-1} u$$

$$y' = \frac{1}{1+u^2}$$

$$\therefore \frac{dy}{dx} = \cos x \times \frac{1}{1+\sin^2 x}$$

$$= \frac{\cos x}{1+\sin^2 x} \quad (2)$$

Question 3 (2 marks)

3(a) (i) $\frac{dT}{dt} = -kCe^{-kt}$ but $\frac{dT}{dt} = -k(T-T_0)$

$\therefore -k(T-T_0) = -kCe^{-kt}$ (1)
 $\therefore T = T_0 + Ce^{-kt}$

(ii) At $t=3$, $T=70$ $T_0 = -20$
 $70 = -20 + 120e^{-3k}$ at $t=0$, $T=100$
 $e^{-3k} = \frac{9}{12}$ $\therefore 100 = -20 + A$
 $\therefore k = \frac{1}{3} \ln \frac{4}{3}$ $\therefore A = 120$ (2)

(iii) let $t=0$, when cup is placed in enviro at 20° .
 $T = 20 + Be^{kt}$
 at $t=0$, $T=70^\circ$
 $70 = 20 + Be^0$
 $\therefore B = 50$

$\therefore T = 20 + 50e^{kt}$

$\therefore T = 20 + 50e^{15(\frac{1}{3} \ln \frac{4}{3})}$
 $\approx 31.86^\circ$
 $\approx 32^\circ \text{C}$ (4)

(b) (i) $P(tz) = 8 - 16 + 8 = 0$ $\therefore$ a zero. (1)

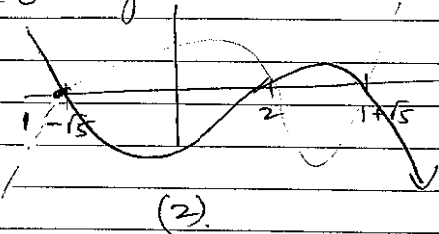
(ii) $x^2 - 2x - 4$
 $\begin{array}{r} x^2 - 2x - 4 \\ x^2 - 4x^2 + 8 \\ -2x^2 \\ (-) -2x^2 + 4x \\ -4x + 8 \\ (-) -4x + 8 \end{array}$
 $\therefore P(x) = (x-2)(x^2 - 2x - 4)$
 $x = \frac{2 \pm \sqrt{4+16}}{2}$
 $= \frac{2 \pm 2\sqrt{5}}{2}$ (2)
 $\therefore x = 2, 1 \pm \sqrt{5}$

b. (iii) $\frac{4}{x-2} \leq x$

$4(x-2) \leq x(x-2)^2$
 $4(x-2) - x(x-2)^2 \leq 0$
 $(x-2)(4 - x^2 + 2x) \leq 0$
 $(x-2)(-x^2 + 2x + 4) \leq 0$ (from (i))

$x=2$ $x = 1 \pm \sqrt{5}$

SS = $\{x: x \geq 1+\sqrt{5}, 1-\sqrt{5} \leq x < 2\}$



Q4 (2 marks)

(b) $(3x^2 + \frac{1}{x})^9$

$$\begin{aligned} T_{R+1} &= {}^9C_R (3x^2)^{9-R} \left(\frac{1}{x}\right)^R \\ &= {}^9C_R \cdot 3^{9-R} x^{18-2R} x^{-R} \\ &= {}^9C_R \cdot 3^{9-R} x^{18-3R} \end{aligned}$$

$\therefore$ coeff of x^3

$$\Rightarrow 18-3R=3$$

$$3R=15$$

$$R=5$$

$$\therefore \text{coeff} = {}^9C_5 \times 3^{9-5}$$

$$= 126 \times 3^4$$

$$= 10206$$

(3)

(c) $f(x) = 1 + e^{2x}$

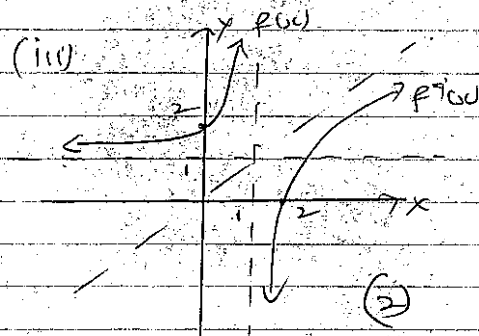
(i) $y > 1$

(ii) $x = 1 + e^{2y}$

$$x-1 = e^{2y}$$

$$\ln(x-1) = 2y$$

$$\therefore y = \frac{1}{2} \ln(x-1)$$



(2)

at $2^{2x+1} - 5(2^x) + 2 = 0$

$$2(2^x) - 5(2^x) + 2 = 0$$

let $u = 2^x$

$$2u^2 - 5u + 2 = 0$$

$$(2u-1)(u-2) = 0$$

$$u = \frac{1}{2}, u = 2$$

$$\therefore x = -1, 1$$

(3)

Q5

(a) (i) $x = 5 \sin \frac{\pi}{2} (t + \frac{1}{3})$

$$\dot{x} = \frac{5\pi}{2} \cos \frac{\pi}{2} (t + \frac{1}{3})$$

$$\ddot{x} = \frac{-5\pi^2}{4} \sin \frac{\pi}{2} (t + \frac{1}{3})$$

$$\therefore \ddot{x} = -\frac{\pi^2}{4} x$$

(2)

(ii) Amplitude = 5, period = $\frac{2\pi}{\pi/2} = 4$ (2)

(b) (i) $\frac{d^2x}{dt^2} = -\frac{72}{x^2}$ at $t=0, x=9, v=4$

$$\frac{d(\frac{1}{2}v^2)}{dx} = \frac{-72}{x^2}$$

$$\frac{1}{2}v^2 = \frac{-72x}{-1} + C_1$$

$$\frac{1}{2}v^2 = \frac{72}{x} + C_1$$

$$x=9 \left. \begin{matrix} v=4 \end{matrix} \right\} \frac{1}{2}(16) = \frac{72}{9} + C_1$$

$$\therefore C_1 = 0$$

$$\therefore \frac{1}{2}v^2 = \frac{72}{x}$$

$$v^2 = \frac{144}{x}$$

$$v = \pm \frac{12}{\sqrt{x}}$$

But particle starts at $t=0$ is travelling to the right at 4m/s $\therefore$ velocity is +ve as $v = \frac{12}{\sqrt{x}}$.

v is never 0 $\Rightarrow \frac{12}{\sqrt{x}} \neq 0$ as $x \neq \infty$ (5)

(ii) $\frac{dx}{dt} = \frac{12}{\sqrt{x}}$

$$\frac{dx}{dt} = \frac{12}{\sqrt{x}}$$

$$t = \frac{x^{3/2}}{18} + C_2$$

at $t=0, x=9$

$$\therefore 0 = \frac{9^{3/2}}{18} + C_2$$

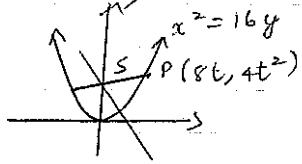
$$\therefore C_2 = -\frac{3}{2}$$

$$\therefore t = \frac{x^{3/2}}{18} - \frac{3}{2}$$

$$t = \frac{x^{3/2}}{18} - \frac{3}{2}$$

(3)

Q6. (12 marks)
(a) (i)



$$\begin{aligned} \text{i) } x^2 &= 16y \\ 2x &= 16 \frac{dy}{dx} \\ \frac{dx}{dy} &= \frac{8}{x} \\ &= \frac{8t}{8} \\ &= t \end{aligned}$$

$$\begin{aligned} \text{Eqn of tan: } y - 4t^2 &= t(x - 8t) \\ y &= tx - 8t^2 + 4t^2 \quad (2) \\ y &= tx - 4t^2 \quad \text{--- (1)} \end{aligned}$$

ii) $S(0,4)$ $P(8t, 4t^2)$

$$\begin{aligned} \text{grad SP} &= \frac{4t^2 - 4}{8t} \\ &= \frac{t^2 - 1}{2t} \end{aligned}$$

$$\therefore \text{grad } l = \frac{-2t}{t^2 - 1}$$

$$\text{eqn: } y - 4 = \frac{-2tx}{t^2 - 1}$$

$$\begin{aligned} t^2 y + t - 4t^2 - y &= -2tx \\ (t^2 - 1)y + 2tx &= 4t^2 - 4 \end{aligned}$$

$$R: (t^2 - 1)y = 2tx = 4(t^2 - 1) \quad \text{--- (2)}$$

iii) sub ① in ②

$$(t^2 - 1)(tx - 4t^2) + 2tx = 4t^2 - 4$$

$$t^3x - 4t^4 - tx + 4t^2 + 2tx = 4t^2 - 4$$

$$t^3x + tx = 4t^4 - 4$$

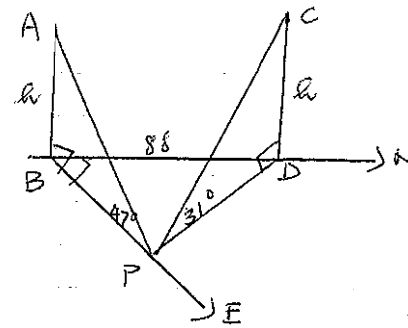
$$x(t^3 + t) = 4t^4 - 4 \quad (1)$$

$$x = \frac{4(t^4 - 1)}{t(t^2 + 1)} = \frac{4(t^2 - 1)(t^2 + 1)}{t(t^2 + 1)} = \frac{4(t^2 - 1)}{t} \quad (3)$$

$$\begin{aligned} \text{sub in ① } y &: t \cdot \frac{4(t^2 - 1)}{t} - 4t^2 \\ &= 4t^2 - 4 - 4t^2 \\ &= -4 \quad (1) \end{aligned}$$

$\therefore$ locus is line $y = -4$

(b)



let height of towers be h .

$$BP = h \cot 47^\circ \quad \checkmark$$

$$PD = h \cot 31^\circ \quad \checkmark$$

In $\triangle BDP$,

$$h^2 \cot^2 31^\circ = 88^2 + h^2 \cot^2 47^\circ$$

$$h^2 (\cot^2 31^\circ - \cot^2 47^\circ) = 88^2$$

$$h^2 = \frac{88^2}{\cot^2 31^\circ - \cot^2 47^\circ} \quad \checkmark$$

$$= \frac{88^2}{\tan^2 59^\circ - \tan^2 43^\circ} \quad \checkmark$$

$$= 4075.2708$$

$$\therefore h = 63.837 \quad \checkmark \quad (5)$$

$\therefore$ height of towers is 64m (to nearest m)

Q 7. (12 marks)

a) Prove $\sum_{r=1}^n r \cdot 2^{r-1} = 1 + (n-1)2^n$

ie prove $1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \dots + n \cdot 2^{n-1} = 1 + (n-1)2^n$

Step 1 Prove true for $n=1$

LHS $= 1 \cdot 2^0 = 1$

RHS $1 + 0 \cdot 2^1 = 1$

$\therefore$ true for $n=1$

(4)

Step 2 Assume true for $n=k$

ie $1 \cdot 2^0 + 2 \cdot 2^1 + \dots + k \cdot 2^{k-1} = 1 + (k-1)2^k$

Step 3 Prove true for $n=k+1$

ie $1 \cdot 2^0 + \dots + k \cdot 2^{k-1} + (k+1)2^k = 1 + k \cdot 2^{k+1}$

LHS $1 + (k-1)2^k + (k+1)2^k$
 $= 1 + k \cdot 2^k - 2^k + k \cdot 2^k + 2^k$
 $= 1 + 2k \cdot 2^k$
 $= 1 + k \cdot 2^1 \cdot 2^k$
 $= 1 + k \cdot 2^{k+1}$
 $= \text{RHS}$

$\therefore$ true for $n=k+1$

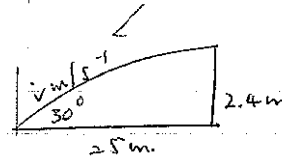
if true for $n=k$

(4)

Step 4 Concl.

We have proved that true for $n=k+1$ if true for $n=k$.
 But that is true for $n=1$ $\therefore$ True for $n=2, 3, 4, n \in \mathbb{N}$

7(b)



(i) Horizontal

$\ddot{x} = 0$

$\dot{x} = c$

$t=0, x = \frac{\sqrt{3}c}{2}$

$\therefore x = \frac{\sqrt{3}c}{2}t$

$x = \frac{\sqrt{3}}{2}vt + c$

when $t=0, x=0$

$x = \frac{\sqrt{3}}{2}vt$

Vertical

$\ddot{y} = -10$

$\dot{y} = -10t + c$

when $t=0, \dot{y} = \frac{v}{2}$

$\dot{y} = -10t + \frac{v}{2}$

$y = -5t^2 + \frac{v}{2}t + c$

when $t=0, y=0$

$\therefore y = -5t^2 + \frac{vt}{2}$ (4)

(i) $t = \frac{2x}{\sqrt{3}v}$

$\therefore y = -5 \left(\frac{2x}{\sqrt{3}v} \right)^2 + \frac{v}{2} \left(\frac{2x}{\sqrt{3}v} \right)$

$y = \frac{-20x^2}{3v^2} + \frac{x\sqrt{3}}{3}$ (1)

ii) when $x = 25, y = 2.4$

$2.4 = \frac{-20 \times 25^2}{3v^2} + \frac{25\sqrt{3}}{3}$

$7.2v^2 = -12500 + 25\sqrt{3}v^2$

$v^2 = \frac{12500}{25\sqrt{3} - 7.2}$

$= 346.246$

$\therefore v = 18.6 \text{ m/s}$

(3)